Accessibility by and to transit; comparison with transit Supply Index results

James Reynolds

2025-08-16

Introduction

Currie and Senbergs [2007] and Currie [2010] previously developed a transit Supply Index (SI) to compare public transport frequency and coverage between across geographic areas (such as census districts). Code to calculate SI scores from GTFS datasets was recently developed as an R package[1], with SI scores[2] also provided as an input to a forthcoming paper on how transit service levels impact housing prices[3].

The SI provides a combined measure of both accessibility 'by transit' and 'to transit' by including service frequency and service coverage in the calculations. However, the extent to which 'by transit' or 'to transit' might influence housing prices is unclear, as the Liang et al. [in press] modeling included only the (combined) SI score. To further explore and compare how housing prices might be influenced by transit accessibility (both 'by' and 'to' transit) a set of corresponding scores need to be calculated.

This note describes the development of these 'by' and 'to' transit scores, as a follow up to the SI scores previously provided.
The rest of this document is structured as follows: the next section outlines the context of the SI. Methods used to develop the ''by' and 'to' transit accessibility metrics are then described, followed by presentation of results for SA1s in Victoria. The results, including differences to the SI scores are then discussed.

[1] See Reynolds et al. [under review] and `https://github.com/James-Reynolds/gtfssupplyindex`

[2] For SA1s accross all of Victoria for the week starting 27/04/23

[3] Liang et al. [in press]

Research context: the Suppy Index

The Supply Index (SI) equation is shown in the margin figure[4], in which: (1) $SI_{area,time}$ is the Supply Index for the area of interest and a given period of time; (2) $Area_{Bn}$ is the buffer area for each stop (n) within the area of interest. In Currie and Senbergs (2007) this was based on a radius of 400 metres for bus and tram stops, and 800

[4] Minor adjustments have been made to generalise the equation, as Currie and Senbergs (2007) focus was the context of Melbourne's Census Collection Districts (CCD) and calculations based on a week of transit service. CCDs predate the introduction of Statistical Areas 1, 2, 3, and 4 (SA1, SA2, SA3, SA4), and other geographical divisions currently used by the Australian Bureau of Statistics (ABS), which may be more familiar to readers.

metres for railway stations; (3) $Area_{area}$ is the area of the area of interest; and (4) $SL_{n,time}$ is the number of transit arrivals for each stop for a given time period.

Methodology

The SI equation lends itself to a simple separation into coverage and frequency components, as per equations 2 and 3. For coverage, this is a measure of how much of each area of interest is within the buffer areas of nearby transit stops. However, as the buffer areas of nearby stops might overlap it is possible to have a coverage value that is larger than 100%.

The frequency measure is a summation of the number of transit arrivals (for the given time period) at stops where there is some overlap of the stop buffer area with the area of interest.

Coverage calculation

As shown in Figure 1, the gtfssupplyindex consists of a number of R functions that perform parts of the SI calculation, starting with: a definition of the buffer distances to be used, a shape file (defining the areas of interest), a gtfs file. These are variously transformed into different tables (e.g. buffer_distance, areas_of_interest, list_gtfs etc.), culminating in the si_by_area table that produces a SI score for each area of interest (or the si_by_area_and_hour table that produces SI scores for each area of intersect for each hour of the day)

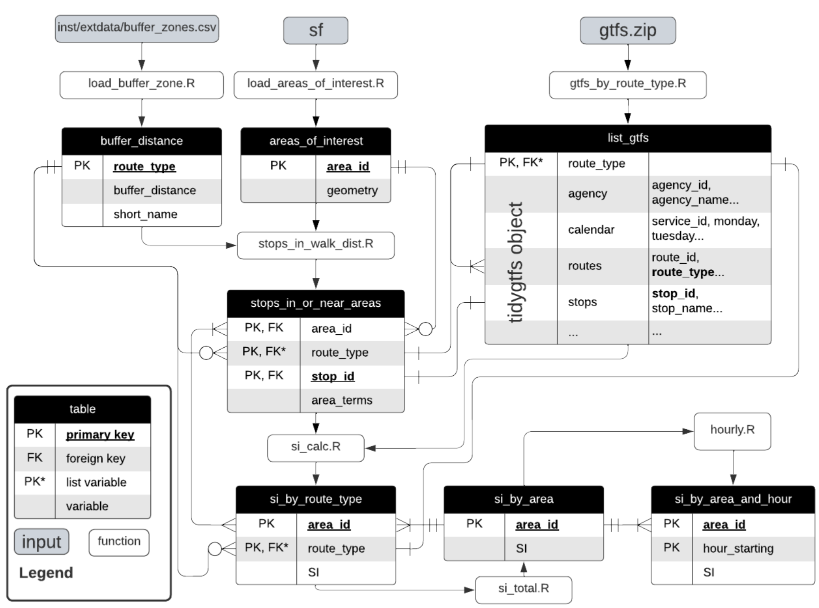

Figure 1: gtfssupplyindex package structure

The stops_in_or_near_areas table is key to calculating coverage, as this table details the area_terms for each combination of area_id, route_type and stop_id. Calculation of the $Coverage_{area}$ value for each area will therefore be a matter of adding the area_terms for all combinations of route_type and stop_id that apply to each area of interest (defined by area_id).

Frequency calculation

There is helper function in the gtfssupplyindex packages, "arrivals", which is called by the si_calc function[5]. This function takes a gtfs file, a list of stop_ids, a date and (optionally) a start and end time and returns a dataframe listing the number of arrivals for each stop_id in the specified time frame[6].

In combination with the stops_in_walk_dist function, this "arrivals" function can be used to calculate the number of transit arrivals at stops that are within or otherwise 'in walking distance' of each area of interest.

[5] The "arrivals" function is not shown in the diagram in Figure 1 as it is a relatively minor function that only performs a small part of the SI calculation.
[6] arrivals <- function(gtfs = gtfs, stop_ids = stop_ids, date_ymd = date_ymd, start_hms = lubridate::hms("-1:0:0"), end_hms = lubridate::hms("48:00:00")){...}

Mornington Penninsula test data

The gtfssupplyindex package includes a set of test data, sourced from the ABS SA1 (2021) boundaries and the Mornington Penninsula Tourist Railway GTFS feed. This GTFS feed includes only three stations and a small number of trips, and so is useful for testing that code runs as expected, and to hand-verify calculation results.

Victoria

Results

Mornington Penninsula

The calculated coverages are shown in Table 1, while Figure 2 shows a plot of the parts of the SA1s that are within the buffer areas around each of the stations.

Table 1: Mornington Penninsula Tourist Railway coverage by SA1

area_id	coverage
21402138114	0.5298114
21402138115	0.0651631
21402138128	0.1514726
21402138502	0.0354876
21402138507	0.0162559

area_id	coverage
21402138508	0.0478445
21402159101	0.7999912
21402159102	0.0168220
21402159105	0.6779951
21402159107	0.6453927
21402159111	0.9776828
21402159112	0.0113615
21402159113	0.1753479
21402159114	0.9531028
21402159115	0.9729829
21402159119	0.8572666
21402159121	0.8591658
21402159122	0.0693544
21402159124	1.0000000
21402159131	1.0000000
21402159134	0.7596671
21402159135	0.1037503
21402159136	1.0000000
21402159202	0.0000451
21402159203	0.4499354
21402159208	0.1885829
21402159210	0.1158566
21402159212	0.0503650
21402159215	0.9882572
21402159221	0.5639974

Results appear consistent with expectations, with most SA1s having only partial coverage. Three SA1s, however, have 100% coverage[7], matching the three small SA1s[8] within the walking catchment of the south-western-most station. The result for area_id 21402159111 also meets expectations, being the SA1 that is almost, but not quite, completely covered by parts of the south-western and centre-most station buffer areas[9].

[7] 21402159134, 21402159131 and 21402159136

[8] Dark blue, light blue and purple

[9] Being 98% covered, in light green

FREQUENCY results for the Mornington Peninsula are shown in Table 2. These also appear to be consistent with expectations, with the areas of interest tending to mostly have 4 or 8 services. This matches with the service pattern for the Mornington Peninsula, which has four trips in each direction on each day that services are run, meaning that the 'middle' station has 8 arrivals while the two terminal stations have 4.

There are, however, some areas of interest that have frequencies of

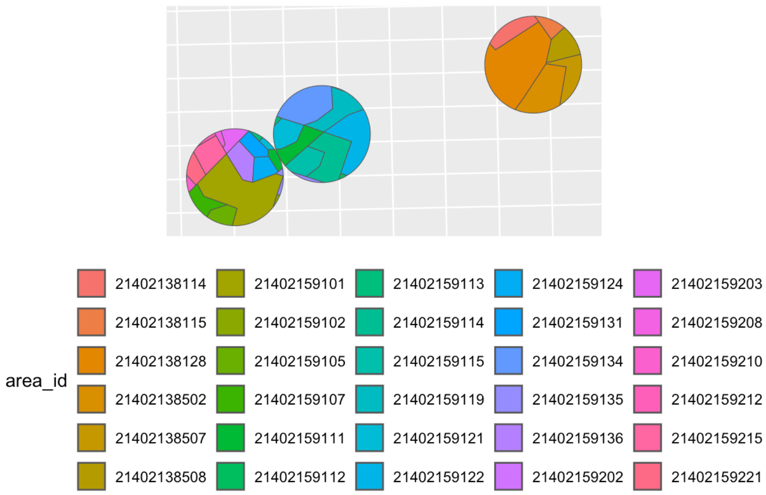

Figure 2: Mornington Penninsula SA1 coverage plot

12. In each case this is because these areas of interest are overlapped by the buffer areas of two of the stations[10]

Table 2: Mornington Penninsula Tourist Railway frequency by SA1

area_id	frequency
21402138114	4
21402138115	4
21402138128	12
21402138502	4
21402138507	4
21402138508	4
21402159101	4
21402159102	4
21402159105	4
21402159107	4
21402159111	12
21402159112	8
21402159113	12
21402159114	8
21402159115	12
21402159119	8
21402159121	12
21402159122	8
21402159124	4
21402159131	4

[10] 21402138128 is in orange in the map above, and covers most of the area of the north-east station, but also a very small sliver of its area within the middle station's buffer area; 21402159111 in green is covered by the south-western and middle stations buffer areas; likewise 21402159113, 21402159115, 21402159121 and 2140259135 in various shades of green, blue and purple.

area_id	frequency
21402159134	8
21402159135	12
21402159136	4
21402159202	4
21402159203	4
21402159208	4
21402159210	4
21402159212	4
21402159215	4
21402159221	4

Comparing the SI scores with the coverage and frequency scores gives the scatterplots and statistical results shown in Figures 3 and 4.

Both show a statistically significant relationship between SI and, respectively, coverage and frequency. The correlation between the calculated values for SI and coverage appears to be better ($r(28) = .89$, $p < .001$ than that between SI and frequency ($r(28) = .39$, $p = .032$). However, this appears to match expectations, given that the test data here (Mornington Penninsula Tourist Railway) has only 4 trips in each direction and the difference between a score of 4, 8 or 12 for frequency might be just a matter of luck as to whether a small piece of the SA1 is within the buffer areas of one or two stations. It might be expected that with more stops and more services (i.e. in the all of Victoria data) that the values for the frequency metric might be more evenly spread.

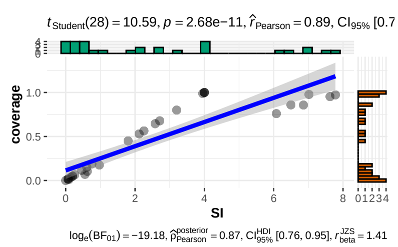

Figure 3: Mornington Penninsula SA1 SI versus coverage

All of Victoria,

```r
mel6_tidy_gtfs <- as_tidygtfs(mel6)
#This fails to return a tidy gtfs

mel6_tidy_gtfs <- mel6 %>% gtfstools::remove_duplicates()
#this dealt with the duplicated agency ids and duplicated calendar_dates entries, but does not the dupl

# find duplicated stops
mel6_tidy_gtfs_duplicated_stops <- tabyl(mel6_tidy_gtfs$stops$stop_id) %>% filter(n>1)
names(mel6_tidy_gtfs_duplicated_stops) <- c("stop_id", "n", "percent")
mel6_tidy_gtfs_duplicated_stops <- left_join(mel6_tidy_gtfs_duplicated_stops, mel6_tidy_gtfs$stops)
## discard duplicate stops
mel6_tidy_gtfs$stops <- mel6_tidy_gtfs$stops[!duplicated(mel6_tidy_gtfs$stops$stop_id),]
```

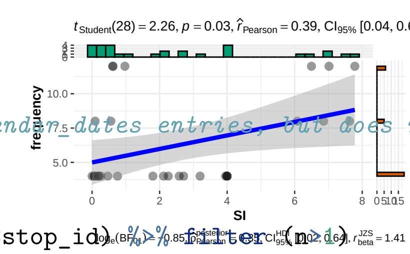

Figure 4: Mornington Penninsula SA1 SI versus frequency

```r
# find duplicated calenders
victoria_gtfs_duplicated_service_ids <- tabyl(victoria_gtfs$calendar$service_id) %>% filter (n>1)
names(victoria_gtfs_duplicated_service_ids) <- c("service_id", "n", "percent")
victoria_gtfs_duplicated_service_ids <- left_join(victoria_gtfs_duplicated_service_ids, victoria_gtfs$ca
## discard duplicate stops
victoria_gtfs$calendar <- victoria_gtfs$calendar[!duplicated(victoria_gtfs$calendar$service_id),]

victoria_gtfs <- victoria_gtfs %>% as_tidygtfs()
#This now returns a tidytransit object.
```

```r
list_gtfs <- gtfs_by_route_type(mel6)

areas_of_interest <- load_areas_of_interest(absmapsdata::sa12021 %>%
            dplyr::filter(state_name_2021 ==  "Victoria") %>%
                                           dplyr::select(sa1_code_2021),
                                          area_id_field = "sa1_code_2021")

buffer_distance <- gtfssupplyindex:::load_buffer_zones()

stops_in_or_near_areas <- gtfssupplyindex:::stops_in_walk_dist(
 list_gtfs = list_gtfs,
 areas_of_interest = areas_of_interest,
 EPSG_for_transform = 28355,
 verbose = FALSE
)

####----run SI.calc function to build si_by_mode_and_time list (by route_type) of tables
```

```r
#victoria_total_SI_SA1_230427_by_day_hour_route_type <- SI_by_day_hour_and_route_type(
```

```r
     list_gtfs = list_gtfs
stops_in_or_near_areas = stops_in_or_near_areas
 start_date_ymd = "2023-04-27"
 end_date_ymd = "2023-05-03"
verbose = TRUE
```

```r
#find first and last arrival times across the list of gtfs
first_arrival <- lapply(list_gtfs, gtfssupplyindex::earliest_arrival) %>% as.data.frame()
first_arrival <- do.call(pmin, first_arrival)
last_arrival <- lapply(list_gtfs, gtfssupplyindex::latest_arrival) %>% as.data.frame()
last_arrival <- do.call(pmax, last_arrival)

#convert into first and last hour (ie. 10, 16) as string
first_hour <- lubridate::hour(first_arrival)
last_hour <- lubridate::hour(last_arrival)

ms <- "00:00"



# create table for output
si_by_day_hour_and_route_type <- data.frame()

# loop SI calculation over days between start_date_ymd and end_date_ymd

dates <- seq(as.Date(start_date_ymd), as.Date(end_date_ymd), by="days")
for(i in as.list(dates)){
  if(verbose){print(paste("Now calculating", i))}
  # create table for output
  si_by_area_route_and_hour <- data.frame()


  # loop SI calculation over hour periods between first and last hour
  for(j in seq(first_hour, last_hour)){
    start_hms <- lubridate::hms(paste(j, ms))
    end_hms <- start_hms + lubridate::hms("1:00:00")
    if(verbose){print(paste("Now calculating", start_hms, "to", end_hms))}
    si_by_route_type <- gtfssupplyindex::si_calc(
      list_gtfs = list_gtfs,
      stops_in_or_near_areas = stops_in_or_near_areas,
      date_ymd = i,
      start_hms = start_hms,
      end_hms = end_hms,
      verbose = verbose)

    #convert list to dataframe
    si_by_route_type <- plyr::ldply(si_by_route_type, data.frame)
    # add hour
```

```
      si_by_route_type$hour_starting <- paste(as.character(lubridate::hour(start_hms)), ":00", sep = "")

      si_by_area_route_and_hour <- rbind(si_by_area_route_and_hour, si_by_route_type)
    }
    si_by_area_route_and_hour$date <- i

    si_by_day_hour_and_route_type  <- rbind(si_by_day_hour_and_route_type, si_by_area_route_and_hour)

  }

  names(si_by_day_hour_and_route_type) <- c(
    "route_type",
    "area_id",
    "SI",
    "hour_starting",
    "date")
```

```
write.csv(victoria_total_SI_SA1_230427_by_day_hour_route_type, "results/victoria_total_SI_SA1_230427_by_


victoria_total_SI_SA1_230427_aggregated <- victoria_total_SI_SA1_230427_by_day_hour_route_type %>%
           group_by(area_id) %>%
           summarize(SI = sum(SI))

write.csv(victoria_total_SI_SA1_230427_aggregated, "results/victoria_total_SI_SA1_230427_aggregated_mel
```

NEXT steps are to calculate the SI for each of the gtfs feeds, then sum to match with the previously calculated versions.

REMEMBER TO CHECK BUFFER DISTANCES….

Discussion and conclusions

References

Graham Currie. Quantifying spatial gaps in public transport supply based on social needs. *Journal of Transport Geography*, 18(1):31–41, 2010. ISSN 0966-6923. DOI: https://doi.org/10.1016/j.jtrangeo.2008.12.002. URL https://www.sciencedirect.com/science/article/pii/S0966692308001518.

Graham Currie and Zed Senbergs. Identifying spatial gaps in public transport provision for socially disadvantaged Australians: the Melbourne needs-gap study. In *Australasian Transport Research Forum*. Australasian Transport Research Forum, 2007.

Jian Liang, Graham Currie, Kang Mo Koo, and James Reynolds. Exploring factors affecting how transit service level influences housing prices by individual transit mode. *Available at SSRN 5144567*, in press.

James Reynolds, Graham Currie, and Yanda Qu. Social needs and gaps in transit service: new gtfs tools & updates to 2021 data. In *Australasian Transport Research Forum 2025*, Auckland, New Zealand, under review.